

ET5218 INNOVATION AND COMMERCIALIZATION OF TECHNOLOGIES

03 Management of New Product Development and Supply Chain

NTU Innovation and
Entrepreneurship

Classification: Restricted

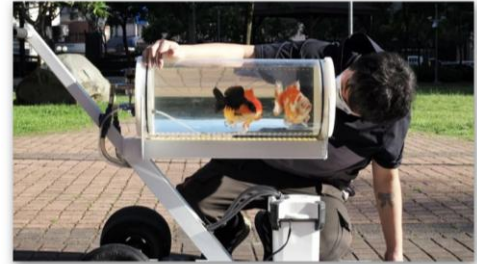


Learning Outcomes

1. Examine the relationship between new products and entrepreneurial prosperity.
2. Identify the different types of New Product Development (NPD).
3. Analyse the impact of Supply Chain.
4. Understand the significance of Disruptive Products and Switching Costs for a new entrepreneurial company.

New Product Development (NPD)

- Developing new products is a part of innovation
- Where innovation becomes real and tangible
- NPD requires significant market intelligence
- Consumer research and competitor research
- It needs careful planning

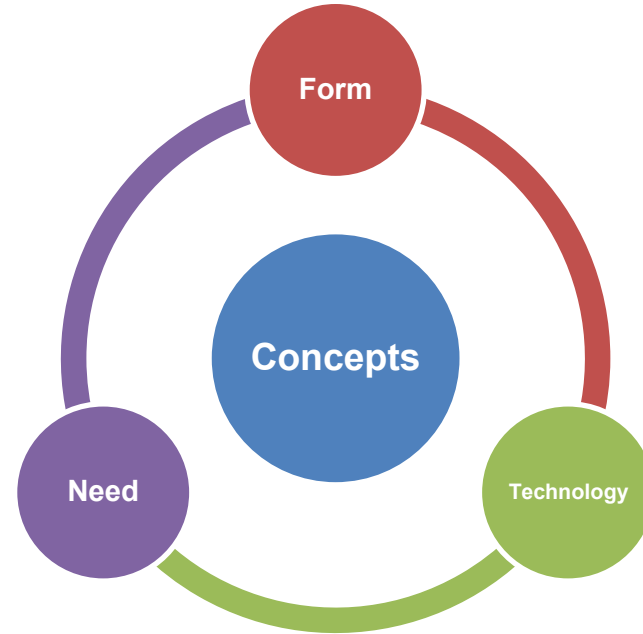


Source: <https://www.dailymail.co.uk/sciencetech/article-10797489/YouTuber-builds-fish-tank-STROLLER-pet-goldfish-walks-town-him.html> accessed on 24 July 2023

New Product Concept

There are three inputs are required to develop New Concepts:

1. Form
2. Technology, and
3. Need



Source: Crawford, M., Di Benedetto, A. (2019). *New Products Management*. McGraw-Hill

Product is Multi-Dimensional



Mature Product Packaging for Product Rejuvenation

Protection and Containment:
Alternate use of

- a) plastic,
- b) aluminum
- c) Treta pak

Identification

- a) Uplifting
- b) More youthful



Source: Pokka.co

NPD as a Growth Strategy

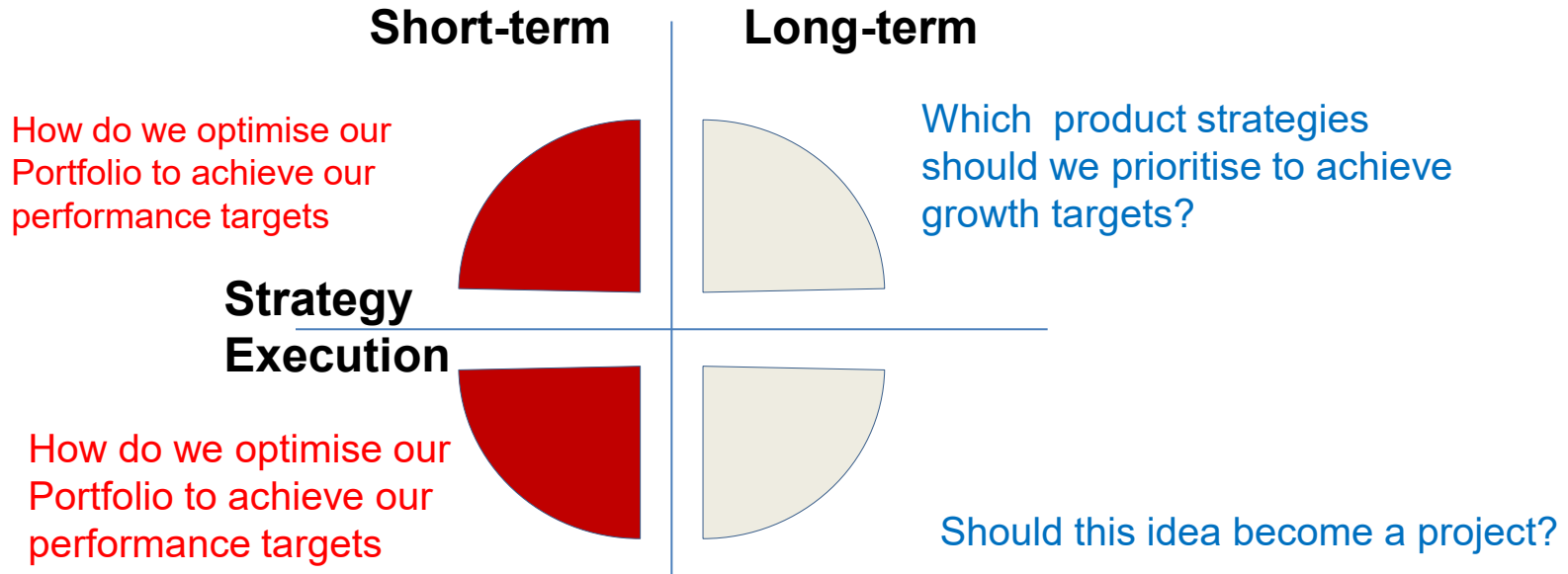
- For Entrepreneur / Corporate Entrepreneur, the area of concern in their growth strategy includes:



Ansoff Matrix

Current Markets	1. Market Penetration Strategy	3. Product Development Strategy
	2. Market Development Strategy	4. Diversification Strategy
New Markets	Current Products	New Products

NPD for Technology Intensive Firms

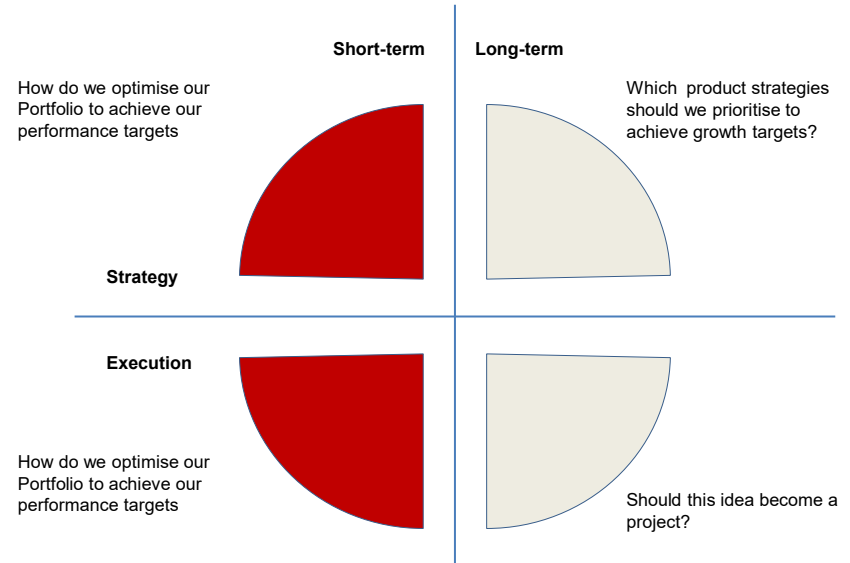
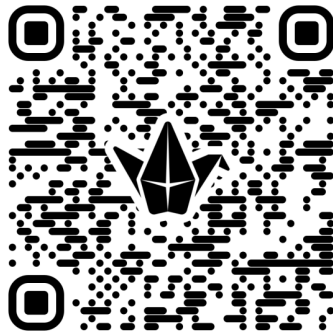


Source: Johnson, S.C. and Jones, C. (1957) How to organise for new products, *Harvard Business Review*, May-June, vol. 35, 49-62.

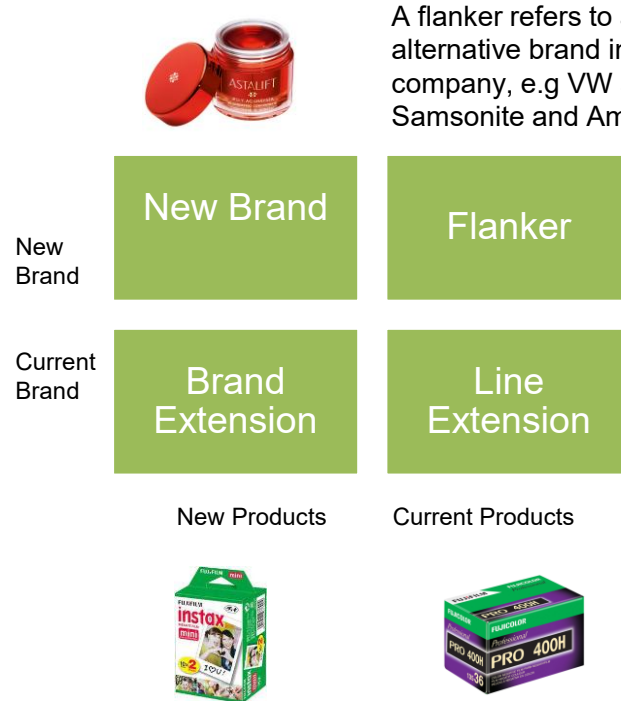
Discussion 3.1

Select an Innovative Product-based Company and identify **success or failure** of the company's

- a) Short-term Strategy & Execution
- b) Long-term Strategy & Execution



Tauber's Growth Matrix



A flanker refers to a lower-priced or alternative brand introduced by a company, e.g VW and Seat or Samsonite and American Tourister

Fujifilm reinvented themselves as Astrafit

Sources:

1. Tauber, E.M. (1981) Brand franchise extension: new product benefits from existing brand names, *Business Horizons*, vol. 24, no. 2, 36–41, with permission from Elsevier.
2. Fujifilm.com

NPD - Design Decisions

- NPD is a sub-process of the innovation process.
- For an entrepreneur or corporate entrepreneur, the five basic decisions to be made are:

What are the target values of the product attributes?

What will the product concept be?

What variants of the product will be offered?

What is the product architecture?

What will be the overall physical form and industrial design of the product?

Dyson developed and is producing ventilators to help treat COVID-19 patients



Source: <https://www.theverge.com/2020/3/26/21195433/dyson-ventilators-covent-coronavirus-develop-produce-uk-nhs-donate-vacuum-motor> accessed on 29 March 20

Source: Krishnan, V., & Ulrich, K. T. (2001). *Product Development Decisions: A Review of the Literature*. *Management Science*, 47(1), 1–21. <http://www.jstor.org/stable/2661556>

Supply Chain

A **supply chain** is the network of organisations, people, activities, information, and resources involved in moving a product or service **from suppliers, manufacturers, distributor, retailer to the final customer.**

Stage	Key Question	Example: Smartphone
 Source	 Where do inputs come from?	 Chips, glass, batteries
 Make	 How is value created?	 Assembly & testing
 Deliver	 How does it reach the market?	 Warehousing & shipping
 Sell	 How does it reach customers?	 Retail stores / online
 Customer	 Who receives the value?	 Smartphone user

Adapted from Source: Chopra, S. (2025). *Supply chain management: Strategy, planning, and operation* (8th ed.). Pearson.

NPD – Supply Chain Decisions



- The following are key questions on decisions that an entrepreneur needs to make surrounding supply chain design:

Who will design and produce the product?

Which new components will be designed specifically for the product?

What is the configuration of the physical supply chain?

What type of process will be used to assemble the product?

Who will develop and supply the process equipment?

Source: Krishnan, V., & Ulrich, K. T. (2001). *Product Development Decisions: A Review of the Literature*. *Management Science*, 47(1), 1–21. <http://www.jstor.org/stable/2661556>

Classification of New Products

New-to-the-world	Undertaken by large firms with substantial technical and marketing resources that produces the first of its kind and create a market.
New product lines (new to the firm)	Large firms playing catch-up by launching and developing their versions by producing products that are not new to the market but new to the company.
Additions to existing lines (line additions)	Large firms playing catch-up by launching and developing their new versions that is significantly different from their current existing versions
Improvements and revisions of existing products	SME developing their own to compete against the originating firm's product and also to replace their existing product line.
Cost Reductions	Cost reduction is the only benefit while the performance of the products remain the same
Repositioning	Discovery of new applications for existing products.

Classification of New Products - Example



Xerox Machine
Xerox Machine Invented by : Chester Carlson
Established in the year : 1938

www.facebook.com/sky.kish

New-to-the-world

New product lines (new to the firm)

Additions to existing lines (line additions)

Improvements and revisions of existing products

Source: Straits Times 5 Apr 2015

PRODUCT EVOLUTION/FUTURE PROOFING

Fine print Canon's inkjet and office printers have kept pace with technological developments to offer cutting-edge features

BUSINESS COPIERS

1965 FIRST PLAIN PAPER COPIER
Canon applied for a patent for its new method of electrophotography, known as XP, which stood for "New Process".
This opened the way to the creation of Canon's first plain paper copier, whose technology changed how office work was performed.
The plain paper copying function enhanced the sharing and communication of information within an organisation as well as work productivity.

1970 GAME CHANGER
Japan launched the first Plain Paper Copier NP-1100. News of the NP system was pivotal to the growth of Canon and as a game changer in its class.
Along with the NP method, Canon patented the use of microcomputer for process control in the NP-1100, which charted the route for modern multifunction devices (MFD) for enterprise use today.

1987 BIRTH OF COLOUR
Canon introduced CLC-1, the first digital full-colour copying machine. It was ahead of its competitors by digitising colour copying machines to enhance work productivity in organisations.
With the enhanced computing capability, the series became part of the everyday workflow in many office environments for document collation and distribution functions with copy, print, fax and scan features.
In the same year, Canon also launched the world's first digital Colour Laser Copier, CLC1. Colour printing in offices grew with business progression. Enterprises produced better documents for business communication and incorporated colour in graphics and pictures.
Colour print also improved work productivity as printed information could be colour-coded to allow for more efficient sorting of information.

2000 DIGITISATION
Canon introduced the imageRunner series network MFD, providing document collation and distribution functions.
Powered with a faster Canon computing architecture, imageChips II, the series was capable of handling various business tasks with exceptional performance, faster processing and support for more sophisticated features.
Full-colour network scanning was introduced and office workers could digitise documents for communication and collaboration.
With the enhanced computing capability, the series became part of the everyday workflow in many office environments for document collation and distribution functions with copy, print, fax and scan features.
In the same year, Canon also launched the world's first digital Colour Laser Copier, CLC1. Colour printing in offices grew with business progression. Enterprises produced better documents for business communication and incorporated colour in graphics and pictures.
Colour print also improved work productivity as printed information could be colour-coded to allow for more efficient sorting of information.

1992 BIRTH OF NETWORKING
The GPRS digital copying machine combined superior image quality with computer capability and heralded the shift to MFD with network capability and facsimile facility.
With an internal computing capability, it utilised parallel processing technology.
The GPRS could function on the network as a printer that could staple, sort and do duplex printing. It also doubled as a digital copier and a fax machine. Having devices connected to the office network greatly boosted work productivity.

2013 ERA OF CLOUD COMPUTING AND MOBILITY
The imageRunner Advance was enhanced with cloud-enabled features for document productivity, security and mobility.
The Scan-to-Cloud feature promised a whole new level of document handling that offer better productivity in office.
Through the Optical Character Recognition engine, scanned documents could be converted into editable formats like Microsoft Word, Excel and PowerPoint for quick and easy use of the information.
The series also had a range of features to help prevent unauthorised information access and use of the MFD to ensure data security. User authentication through proximity cards or pin code also effectively managed an organisation's printing cost.
The proliferation of mobile devices had led to office workflow increasing need for printing solutions that are accessible anytime and anywhere within the office.
With Canon Secure Mobile Print, documents could be printed directly from mobile devices.

2015 ALL NEW
The new generation imageRunner Advance launches.

INKJET

1985 INTRODUCING BUBBLE JET
Canon B-jet was the world's first inkjet printer using bubble jet technology, where a microheater is built into each ink nozzle of a print head.
Ink was heated by running an instantaneous electric current through the microheater so that ink droplets jetted out of the nozzle under the pressure of the heat.
Allowing for affordable and high-quality prints, this technology enabled users to print photos from the comfort of their homes.

2010 RECORDED ENVIRONMENTAL FRIENDLY
Canon launched the imageRunner Advance series, the next-generation office network MFD with advanced intuitive user interface and collaborative information technology (IT) environment integration.
Certified with the Singapore Green Label, the lead-free and mercury-free.

1999 TAKE IT EVERYWHERE
The world's smallest and first portable printer B-10 series measured 31cm by 25cm by 4.8cm.

1993 AFFORDABLE INK
Canon developed the first individual ThinkTank ink cartridge system combining optical detection, dot count method and visual inspection for greater cost-efficiency in the BJC-600 printer series.
This was different from previous technology as it featured a new ink cartridge design with separate ink tanks for each colour. Users could replace individual ink cartridges, rather than the entire cartridge, when it runs out. This thereby optimised cost savings.

1999 GROUNDBREAKING INVENTION
Canon created the world's first nozzle production method termed as Fine Full-Photolithography Inkjet Nozzle Engineering with BJC-8000. This produced outstanding photo quality.

2011 MORE SPECIFIC COLOUR RANGE
Canon's first desktop Professional Inkjet Printer with 12-colour ink system was the Pixma P915-L, which produces A3 size prints.
The Lucia pigment ink system expanded the colour gamut with improved saturation of colours and darker, deeper blacks with its five monochrome inks.
With the introduction of Pixma P915-L, Canon was able to bring the digital darkroom to the photographer, realising simple and affordable professional photo printing at home.

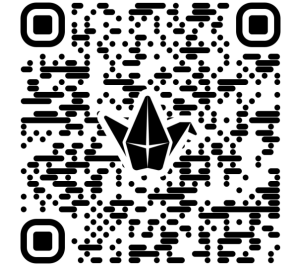
2015 PRINT AND SCAN ON THE GO
Flagship wireless home photo all-in-one Pixma MG7570 comes with near field communication technology, Pixma Cloud Link and mobile connectivity with Pixma Printing Solutions App.
With Pixma Printing Solutions App, users will enjoy a seamless printing experience in terms of wireless printing and scanning on the go, as well as cloud printing service, through its integration across various mobile platforms and cloud storage.

— Compiled by Lai Yi Ming

Cost Reductions

Repositioning

Discussion 3.2



New-to-the-world

New product lines (new to the firm)

Additions to existing lines (line additions)

Improvements and revisions of existing products

Cost Reductions

Repositioning

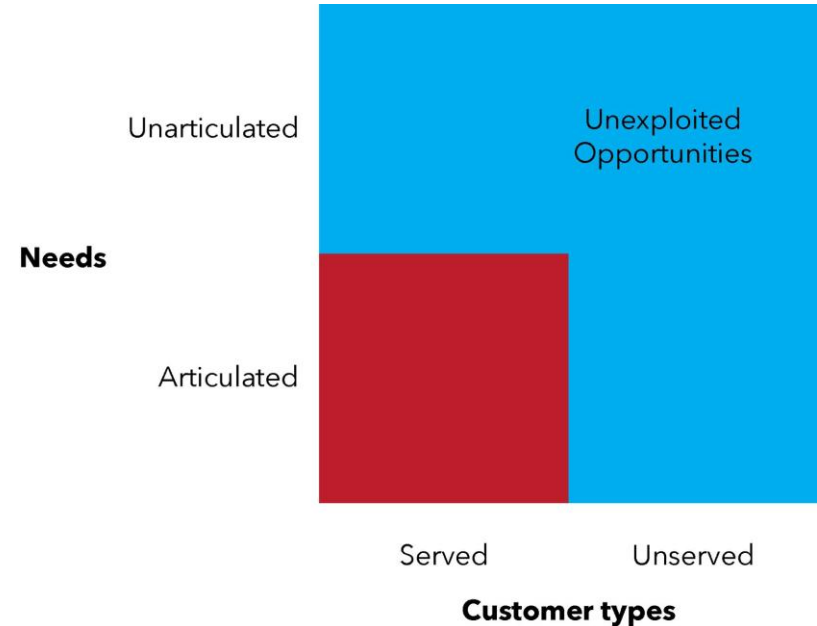
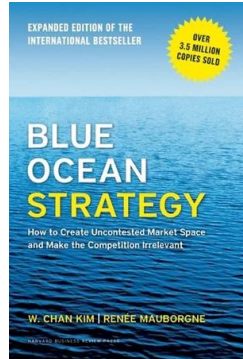
Provide examples
for Classification of
New Products

Market Research

- Lead by Sales and Marketing
- Identify new markets and products
- Predicting success patterns & obtaining customer reviews
- Identifying barriers to success
- Improvements to existing products
- Developing new features
- Reducing risk for the organisation

Market Research and Gaining Future Customers

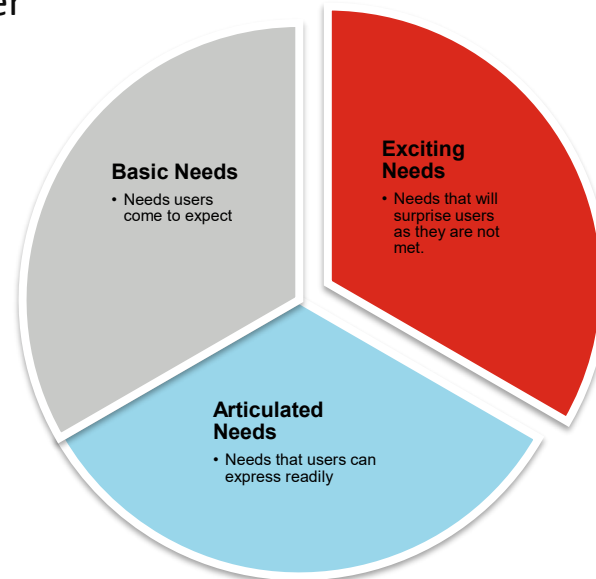
Market Research is to effectively uncover customer needs.



Source: Needs and Customer Types [Adapted from Trott, P. (2021)]

Types of Human Needs

According to King (1985), there are 3 types of user needs.



Source: King, L. J. (1985). A model of resources and human needs. *Environmental Conservation*, 12(3), 249–253. <https://doi.org/10.1017/S0376892900015988>



Source:
https://www.reddit.com/r/singapore/comments/nj0ml6/grab_driver_going_the_extra_mile_with/ accessed on 16 May 2025

Needs, Wants and Desires

Needs...



Wants

- Not immediate needs
- Not permanent
- Changes with time
- Users spend time to select the services; time and delivery of service



Desires

- User wants turn to willingness to use & pay for a service
- Creates a demand for a service



NPD Models

1. Departmental stage models
2. Activity stage models
3. Cross functional models & concurrent engineering
4. Decision-stage models
5. Conversion process models
6. Response models
7. Network models
8. Outsourced models

NPD Models (Traditional)

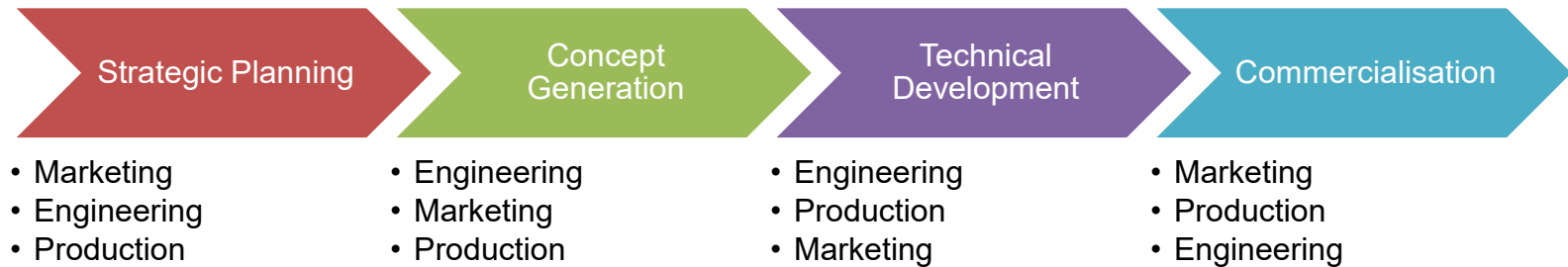


Linear NPD Model

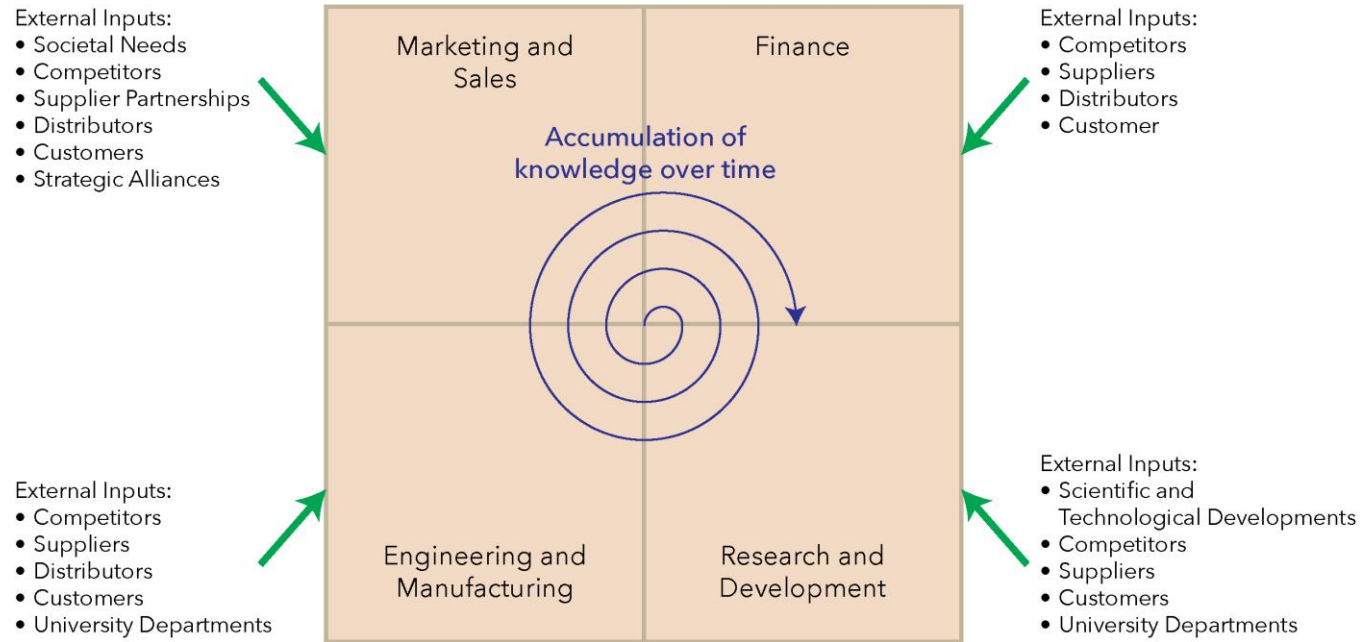
- Departmental stage models, Activity stage models, Decision-stage models
- Each stage needs to be completed before moving onto the next
- Painful slow to bring idea to the market



NPD Model – Concurrent Engineering



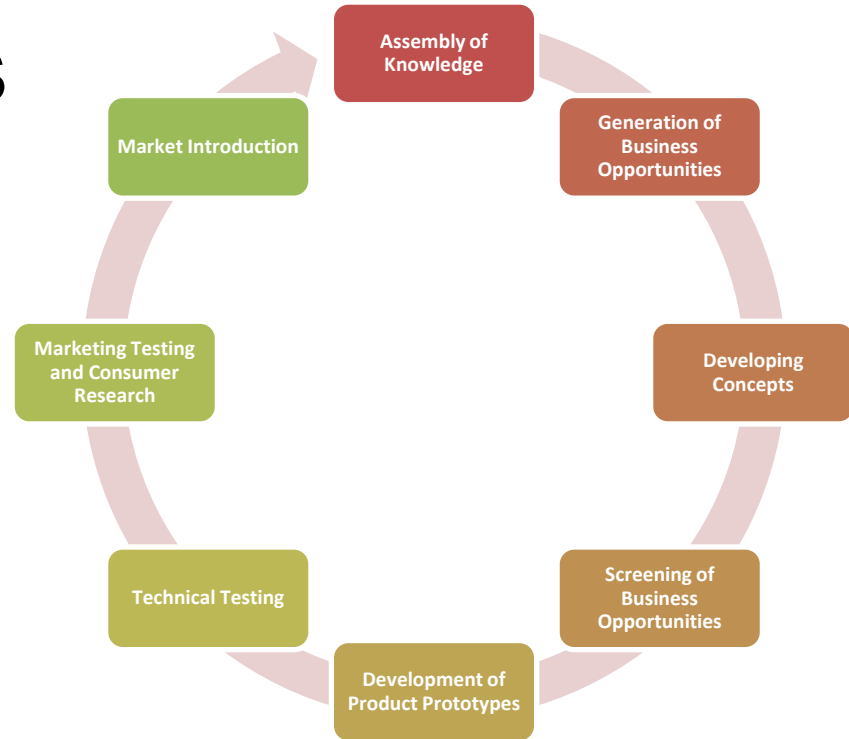
NPD Model – Network Model



Adapted from Source: Trott, P. (2021).
Innovation Management and New Product
Development. 7th edition. Pearson.

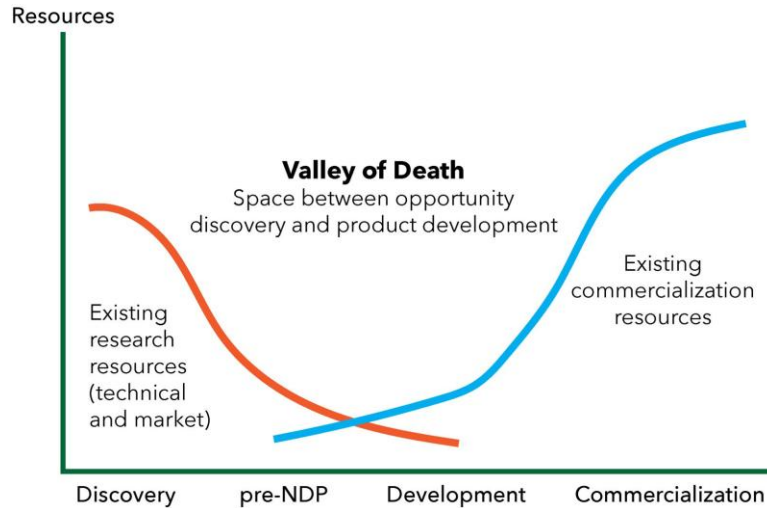
Key NPD Activities

NPD process is a series of linked activities



Adapted from Source: Trott, P. (2021). Innovation Management and New Product Development. 7th edition. Pearson.

Valley of Death

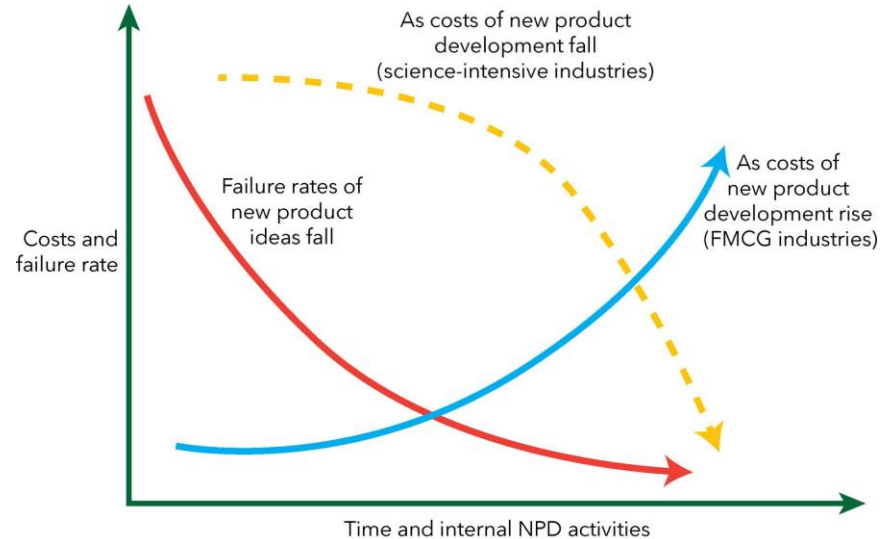


Adapted from Source: Markham, S., Ward, S., Aiman-Smith, L., Kingon, A.. (2010) *The Valley of Death as Context for Role theory in Product Innovation*

- “Valley of Death” describes that period in the development of a product or service when a significant increase in investment is required, making the risk of failure much more likely to outweigh any potential future return.
- This phase is typically occurring after the company launches a product but has not yet seen any revenue.
- The term comes from plotting the shape of a company’s cash flow onto a graph.

Attrition of New Products

- The failure rates of ideas drop as the NPD activities march towards development maturity.
- the costs of science-intensive NPD fall over time
- the costs of Fast Moving Consumer Goods (FMCG) products increase towards the end of development



Adapted from Source: Trott, P. (2021). Innovation Management and New Product Development. 7th edition. Pearson.

Reasons for New Product Failure

Nothing New

Inadequate Budget for Development

Poor Market Research, Positioning

Lack of Management Support

Did not involve the Customer

etc

Discussion 3.3

- Examples of Product Failures

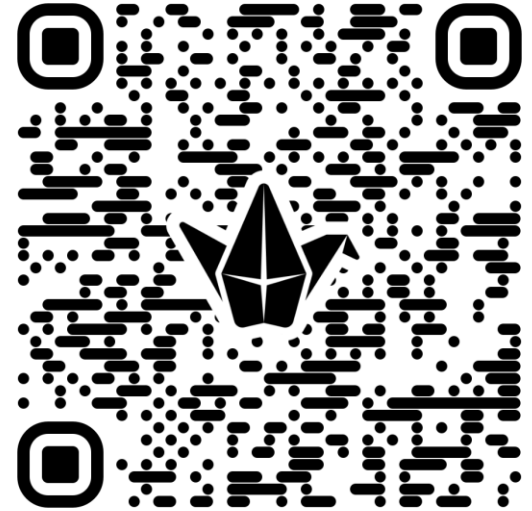
Nothing New

Inadequate Budget for Development

Poor Market Research, Positioning

Lack of Management Support

Did not involve the Customer



Future Product Trends

Future Product Trend	What It Means	Example
 AI-Enabled Products	 Products become smarter, predictive and responsive to users.	 AI assistants, smart appliances
 Hyper-Personalisation	 Products and experiences are increasingly customised using customer data and AI.	 Spotify/Netflix recommendations
 Sustainable & Circular Products	 Greater emphasis on lower environmental impact, reuse, repair and recycling.	 Refillable packaging, recycled materials
 Connected Products (IoT)	 Physical products connect to networks, apps and other devices.	 Smart watches, connected cars
 Immersive Experiences	 AR/VR and spatial technologies blend physical and digital product experiences.	 Virtual product demonstrations
 Product-as-a-Service	 Customers increasingly pay for access or outcomes, rather than simply owning products.	 Software subscriptions, equipment leasing
 Key Takeaway: Future products will increasingly be smart, personalised, connected, sustainable and service-oriented.		

Sources: Gartner (2025) Top Strategic Technology Trends for 2025; Deloitte (2025) 2025 Connected Consumer Survey; World Economic Forum (2024) Circular Transformation of Industries: Unlocking Economic Value.

Adapted from Sources:

1. Gartner. (2025). *Top strategic technology trends for 2025*.
2. Deloitte. (2025). *2025 connected consumer survey*.
3. World Economic Forum. (2024). *Circular transformation of industries: Unlocking economic value*.

End